

Advanced CPU Architecture & Hardware Accelerators

LAB4 – FPGA, Quartus, Timing & Power Analysis, and Validation

by

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1. Introduction and Aim

This report documents the synthesis, timing, power and area (PPA) analysis, and functional validation of the digital system designed for LAB4. The objective is to synthesize a digital system that combines a combinational subpart (the ALU) and a synchronous subpart (the PWM output unit) onto an FPGA target, and to characterize its Performance, Power, and Area (logic usage).

As required by the assignment, a pure combinational block (the ALU) cannot be timed on its own, because static timing analysis requires registers to confine the logic paths. Therefore, when analyzing the ALU, the device under test is confined between an input register and an output register driven by the same clock, as shown in the test-case diagram of the assignment. For the synchronous subpart and the full system this requirement is met inherently, since they already contain registers clocked by the system clock.

1.1 System overview

The top-level digital system is structural and is composed of two concurrent subparts:

- Combinational Digital Circuit – the ALU. It receives the operands X and Y and the function selector ALUFN, and produces the result ALUout together with the four status flags N (Negative), C (Carry), Z (Zero) and V (Overflow). It implements the arithmetic, shift and boolean instructions of the ISA.
- Synchronous Digital Circuit – the PWM output unit. It contains a 16-bit free-running timer, a comparator against the X and Y threshold values, and mode-dependent output logic (Set/Reset, Reset/Set, Toggle) producing the PWMout signal. It is driven by CLK, ENA and RST and is selected by the lower ALUFN bits (PWM mode).

2. Functional Validation – ModelSim Simulation

Before synthesis, the complete `digital_user_interface` (the top entity including the I/O wrapper, the PLL, the ALU and the synchronous PWM unit) was simulated in ModelSim with maximal coverage. The two waveforms below zoom in on the regions of interest of that simulation. Irrelevant signals were removed and the signals were grouped so that the timer, the PWM thresholds, the ALU outputs, the flags and the board I/O can be followed together.

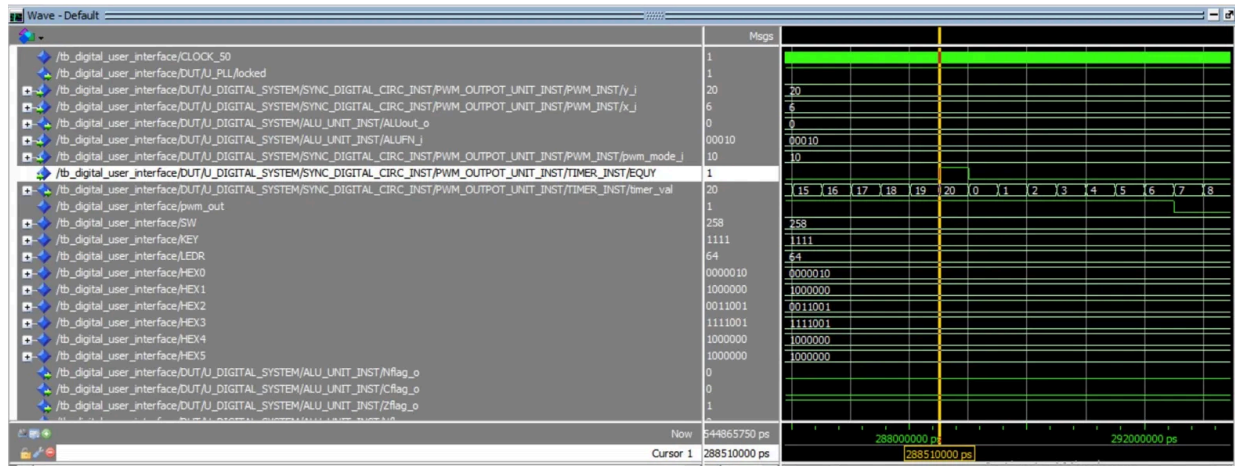


Figure 1: ModelSim simulation of `digital_user_interface` – region of interest 1: the 16-bit timer (`timer_val`) rolls over at the maximum count, the `EQUY` pulse is generated when the timer matches the Y threshold, and `PWM_out` toggles accordingly. The PLL locked signal is high, confirming a stable working clock.

In the first region of interest the free-running timer value (`timer_val`) is seen counting up through 15, 16, 17, 18, 19, 20 and then wrapping to 0, 1, 2 and so on. The `y_i` threshold is 20 and the `x_i` threshold is 6, while `pwm_mode_i` = 10 (Toggle mode). When the timer reaches the Y value the `EQUY` (equal-to-Y) signal pulses, which is exactly the event that drives the PWM output logic. also we can see that the `pwm_out` is going up when timer geting to 6 - the x value

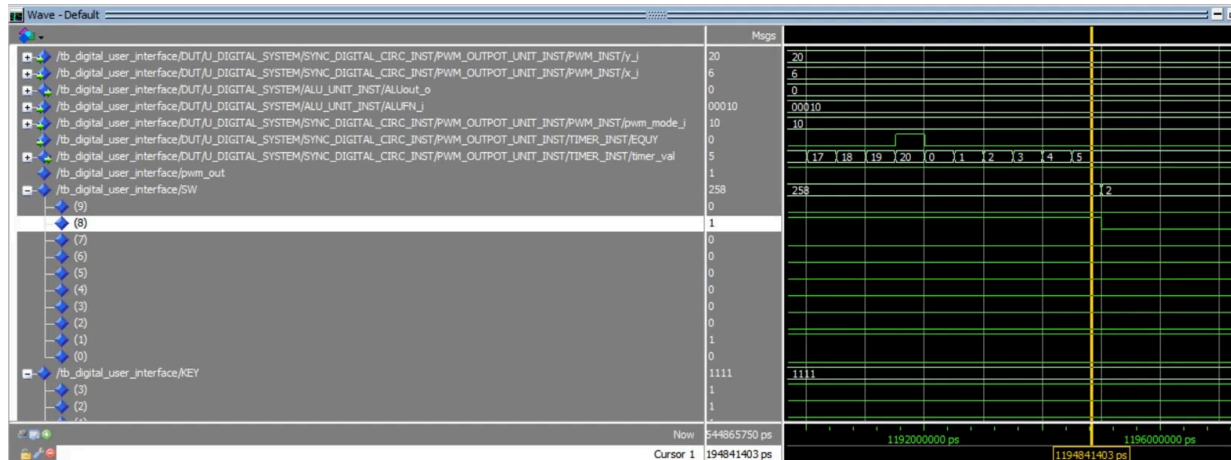


Figure 2: ModelSim simulation of digital_user_interface – region of interest 2: detailed view of the timer wrap and the EQUY pulse, with the individual SW bits expanded. When SW changes (bit 8 toggles, SW becomes 2) the operating point of the PWM unit changes accordingly.

The second region of interest expands the individual switch bits and zooms further into the timer wrap. The EQUY pulse is visible exactly when timer_val passes the Y threshold (20), and pwm_out reacts on that edge. The switch vector SW is shown bit-by-bit; a change of SW (here from 258 to 2, with bit 8 toggling) changes the input applied to the PWM/ALU and demonstrates that the interface registers capture the new value correctly. Together the two captures confirm that the timer, the comparison logic, the PWM mode control and the I/O path all behave as specified by the ISA before the design is committed to synthesis.

3. Combinational Subpart – ALU

The ALU is a pure combinational block. To allow timing analysis it is wrapped between an input register and an output register clocked by a single clock CLK (entity ALU_unit_Wrapper), as required by the assignment. The following results were produced from the Quartus compilation and timing/power reports of this wrapper.

3.1 Area (logic usage)

Flow Summary	
<<Filter>>	
Flow Status	Successful - Sat May 30 13:29:49 2026
Quartus Prime Version	21.1.0 Build 842 10/21/2021 SJ Lite Edition
Revision Name	TEST
Top-level Entity Name	ALU_unit_Wrapper
Family	Cyclone V
Device	5CSXFC6D6F31C6
Timing Models	Final
Logic utilization (in ALMs)	82 / 41,910 (< 1 %)
Total registers	33
Total pins	35 / 499 (7 %)
Total virtual pins	0
Total block memory bits	0 / 5,662,720 (0 %)
Total DSP Blocks	0 / 112 (0 %)
Total HSSI RX PCSs	0 / 9 (0 %)
Total HSSI PMA RX Deserializers	0 / 9 (0 %)
Total HSSI TX PCSs	0 / 9 (0 %)
Total HSSI PMA TX Serializers	0 / 9 (0 %)
Total PLLs	0 / 15 (0 %)
Total DLLs	0 / 4 (0 %)

Figure 3: ALU – Flow Summary (area / resource usage report).

The synthesis report shows that the ALU is very small relative to the device. It uses 82 of 41,910 ALMs (< 1%), 33 registers (these come from the input/output wrapper registers used for timing), and 35 of 499 pins (7%). No block memory, DSP blocks, PLLs or DLLs are used, which is expected for a purely combinational arithmetic/logic block.

Resource	Usage
Logic utilization (ALMs)	82 / 41,910 (< 1%)
Total registers	33 (wrapper I/O registers)
Total pins	35 / 499 (7%)
Block memory bits	0
DSP blocks	0
PLLs / DLLs	0 / 0

3.2 Maximum frequency (Fmax)

	Fmax	Restricted Fmax	Clock Name	Note
1	155.59 MHz	155.59 MHz	CLK	

Figure 4: ALU – maximum operating frequency reported by the Timing Analyzer.

With the wrapper registers in place, the Timing Analyzer reports a maximum frequency of 155.59 MHz on the CLK domain (restricted Fmax 155.59 MHz). This represents the speed at which the combinational ALU logic, captured between two registers, can be clocked. The ALU is the slower of the two subparts because its longest combinational path (the arithmetic/shift datapath) is longer than the per-stage logic of the synchronous PWM unit.

3.3 Critical path

Path #1: Setup slack is 13.573		
Path Summary	Statistics	Data Path
Property	Value	
1 From Node	ALUFN_reg[4]	
2 To Node	ALUout_o[7]~reg0	
3 Launch Clock	CLK	
4 Latch Clock	CLK	
5 Data Arrival Time	10.236	
6 Data Required Time	23.809	
7 Slack	13.573	

Figure 5: ALU – critical path summary (setup), slack 13.573 ns.

The critical (longest) path of the ALU runs from the function-selector register ALUFN_reg[4] to the output register ALUout_o[7]~reg0. The data arrival time is 10.236 ns against a data-required time of 23.809 ns, giving a positive setup slack of 13.573 ns. The path passes through the function decode and the shifter/arithmetic datapath before reaching the most-significant output bit, which is why this bit determines the ALU's maximum frequency.

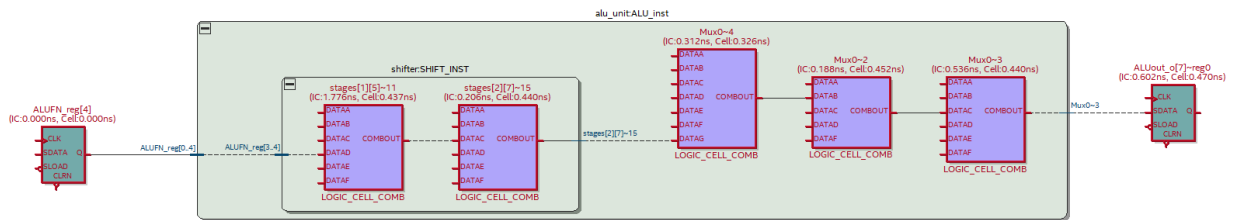


Figure 6: ALU – Technology-Map / fitter view of the critical path: ALUFN register → shifter stages → multiplexers → output register.

The technology-map view of the path shows the source register (ALUFN), the shifter stages (stage[1] and stage[2] logic cells), the chain of output multiplexers (Mux0~1 ... Mux0~4) that select between the arithmetic, shift, logic and boolean results, and finally the output register ALUout_o[7]. The selection multiplexer chain at the output is the part that contributes most of the interconnect/cell delay on this path.

3.4 Chip planner (placement)

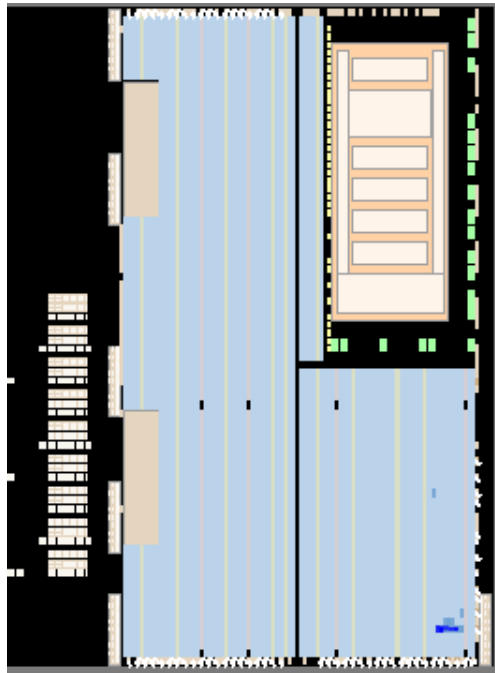


Figure 7: ALU – Chip Planner placement of the wrapped ALU on the Cyclone V die.

The Chip Planner view shows that, because the design is tiny, the placed-and-routed logic occupies only a small, compact region of the device fabric, with the I/O pads spread along the device edges.

3.5 RTL and clock verification

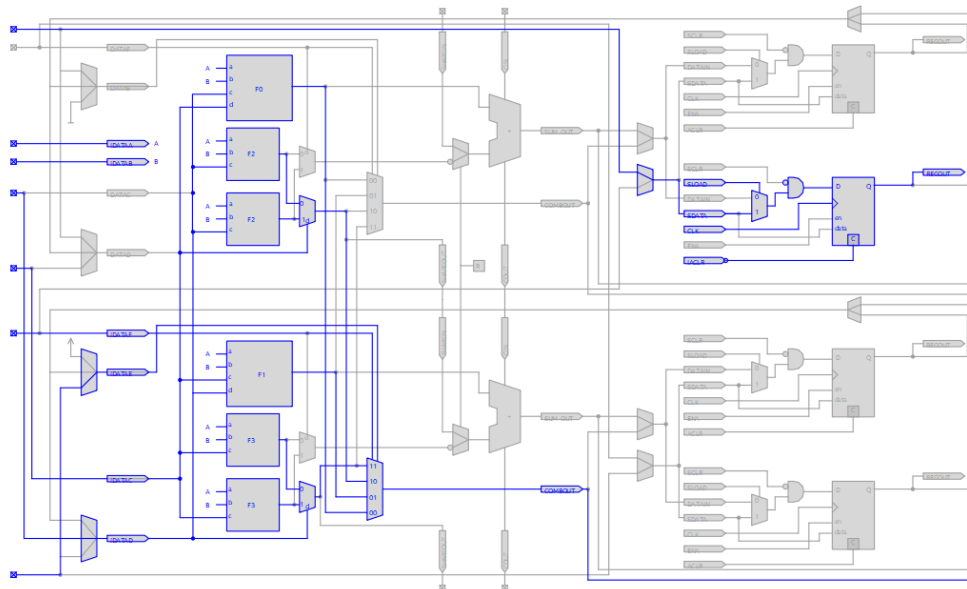


Figure 8: ALU – RTL Viewer of the combinational ALU internals (adder/subtractor, logic, shifter and the output selection).

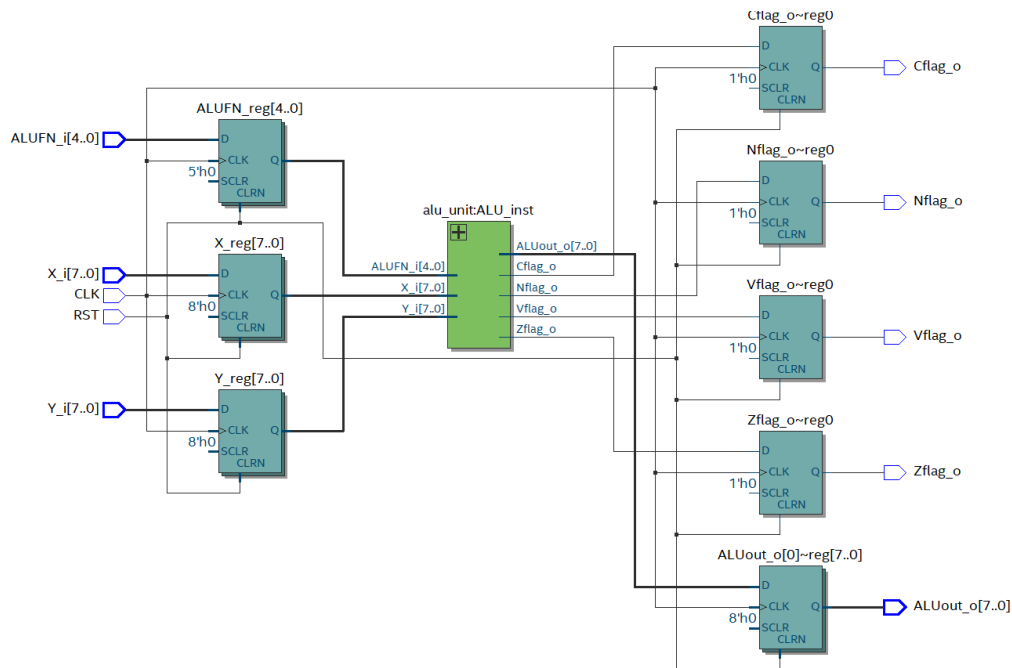


Figure 9: ALU wrapper – RTL Viewer showing the input registers (ALUFN_reg, X_reg, Y_reg), the alu_unit instance, and the output registers for ALUout and the N/C/Z/V flags, all driven by CLK.

The RTL Viewer confirms the structure used for timing: the operands X and Y, together with ALUFN, are first captured in the input registers (8-bit X_reg and Y_reg, 5-bit ALUFN_reg), feed the alu_unit:ALU_inst combinational core, and the result ALUout_o[7..0] and the four flags Cflag, Nflag, Vflag, Zflag are then re-registered before leaving the wrapper. This matches the “Reg → ALU → Reg” test-case from the assignment.

Clock Name	Type	Period	Frequency	Rise	Fall	Duty Cycle	Divide by	Multiply by	Phase	Offset	Edge List	Edge Shift	Inverted	Master	Source	Targets
CLK	Base	20.000	50.0 MHz	0.000	10.000											{ CLK }

Figure 10: Clock constraint used for the ALU wrapper: CLK, base clock, 20 ns period (50 MHz nominal) with a 10 ns rise time.

The clock used to constrain the wrapper is a base clock CLK with a 20 ns period (50 MHz). Constraining the clock this way lets the Timing Analyzer report the true Fmax of the confined logic shown above.

4. Synchronous Subpart – PWM Output Unit

The synchronous subpart (sync_digital_circ) contains the 16-bit timer, the comparison logic against the X and Y thresholds, and the mode-dependent PWM output control. Because it is already a clocked design, no wrapper is needed for timing analysis. The reports below characterize this subpart on its own.

4.1 Area (logic usage)

Flow Summary	
<<Filter>>	
Flow Status	Successful - Sun May 31 19:46:17 2026
Quartus Prime Version	21.1.0 Build 842 10/21/2021 SJ Lite Edition
Revision Name	TEST
Top-level Entity Name	sync_digital_circ
Family	Cyclone V
Device	5CSXFC6D6F31C6
Timing Models	Final
Logic utilization (in ALMs)	54 / 41,910 (< 1 %)
Total registers	49
Total pins	41 / 499 (8 %)
Total virtual pins	0
Total block memory bits	0 / 5,662,720 (0 %)
Total DSP Blocks	0 / 112 (0 %)
Total HSSI RX PCSs	0 / 9 (0 %)
Total HSSI PMA RX Deserializers	0 / 9 (0 %)
Total HSSI TX PCSs	0 / 9 (0 %)
Total HSSI PMA TX Serializers	0 / 9 (0 %)
Total PLLs	0 / 15 (0 %)
Total DLLs	0 / 4 (0 %)

Figure 11: PWM / synchronous subpart – Flow Summary (area).

The synchronous subpart uses 54 of 41,910 ALMs (< 1%), 49 registers (the 16-bit timer plus the PWM control registers), and 41 of 499 pins (8%). It uses no block memory, DSP or PLL resources — the PLL belongs to the top level, not to this subpart.

Resource	Usage
Logic utilization (ALMs)	54 / 41,910 (< 1%)
Total registers	49
Total pins	41 / 499 (8%)
Block memory bits	0
DSP blocks / PLLs	0 / 0

4.2 Maximum frequency (Fmax)

	Fmax	Restricted Fmax	Clock Name	Note
1	223.21 MHz	223.21 MHz	clk_i	

Figure 12: PWM subpart – maximum frequency on the clk_i domain.

The Timing Analyzer reports a maximum frequency of 223.21 MHz for the synchronous subpart on its clk_i domain. This is considerably higher than the ALU's 155.59 MHz, because the logic between two consecutive timer registers is short (an increment and a compare) compared with the ALU's full arithmetic/shift datapath.

4.3 Critical path

Path #1: Setup slack is 15.520				
Path Summary	Statistics	Data Path	Waveform	Extra Fitter Information
Property	Value			
1 From Node	pwm_output_unit:PWM_OUTP...er:TIMER_INST count_q[0]			
2 To Node	pwm_output_unit:PWM_OUTP...er:TIMER_INST count_q[2]			
3 Launch Clock	clk_i			
4 Latch Clock	clk_i			
5 Data Arrival Time	8.277			
6 Data Required Time	23.797			
7 Slack	15.520			

Figure 13: PWM subpart – critical path summary (setup), slack 15.520 ns.

The critical path of the subpart runs internally inside the timer, from TIMER_INST|count_q[0] to TIMER_INST|count_q[2], both clocked by clk_i. The data arrival time is 8.277 ns against a required time of 23.797 ns, giving a setup slack of 15.520 ns. The path therefore lies in the increment/carry logic of the 16-bit counter.

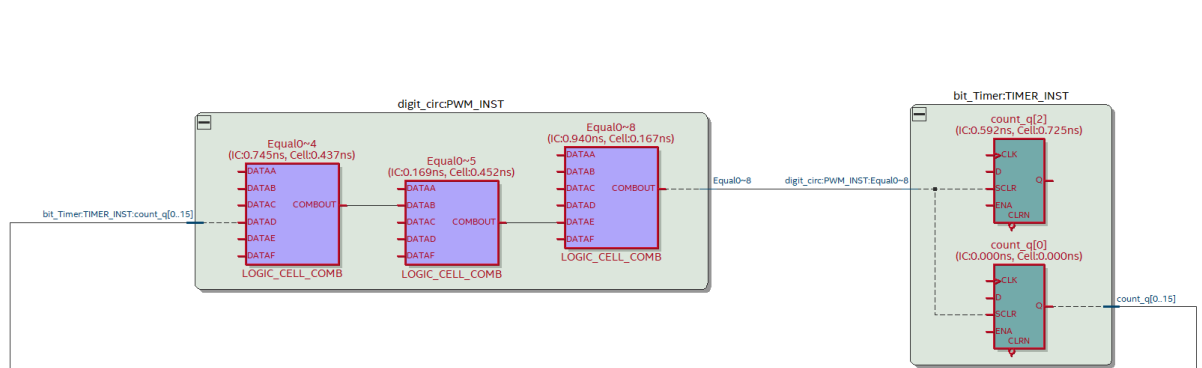


Figure 14: PWM subpart – Technology-Map view: the timer registers (count_q) and the comparison logic cells (Equal0~4/5/8) that feed the PWM output.

The technology-map view shows the timer registers count_q feeding the comparator logic cells Equal0~4, Equal0~5 and Equal0~8, which together implement the comparison of the timer value against the X and Y thresholds. The result of this comparison drives the PWM output control logic.

4.4 Chip planner (placement)

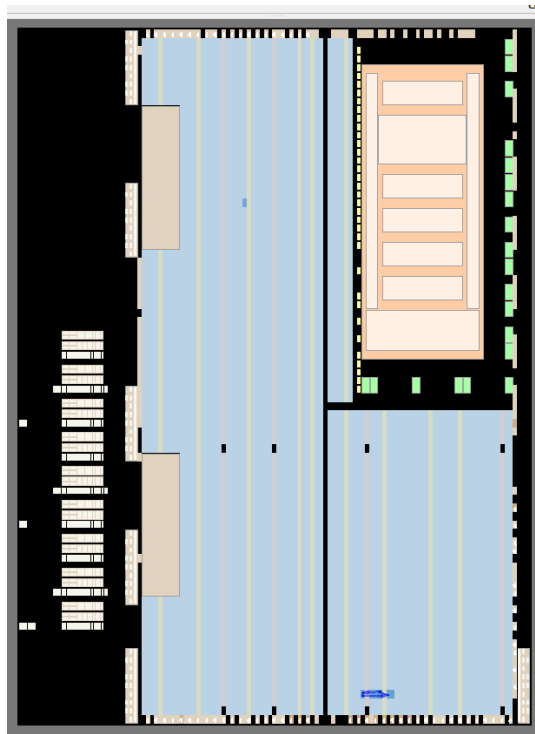


Figure 15: PWM subpart – Chip Planner placement.

4.5 Power

Power Analyzer Summary	
<<Filter>>	
Power Analyzer Status	Successful - Sun May 31 19:41:02 2026
Quartus Prime Version	21.1.0 Build 842 10/21/2021 SJ Lite Edition
Revision Name	TEST
Top-level Entity Name	sync_digital_circ
Family	Cyclone V
Device	5CSXFC6D6F31C6
Power Models	Final
Total Thermal Power Dissipation	422.12 mW
Core Dynamic Thermal Power Dissipation	0.00 mW
Core Static Thermal Power Dissipation	411.24 mW
I/O Thermal Power Dissipation	10.88 mW
Power Estimation Confidence	Low: user provided insufficient toggle rate data

Figure 16: PWM subpart – Power Analyzer summary.

The Power Analyzer estimates a total thermal power of 422.12 mW for the synchronous subpart. This is dominated by the static (leakage) component of 411.24 mW, with essentially no core dynamic power (0.00 mW) and 10.88 mW of I/O thermal power. The confidence is reported as Low because insufficient toggle-rate (switching activity) data was supplied; with a real .vcd from the simulation the dynamic estimate would be more accurate. The static power being dominant is typical for a small design on a large device, where leakage of the whole die outweighs the activity of a few hundred logic cells.

Power component	Value
Total thermal power dissipation	422.12 mW
Core dynamic thermal power	0.00 mW
Core static thermal power	411.24 mW
I/O thermal power	10.88 mW
Estimation confidence	Low (insufficient toggle-rate data)

4.6 RTL

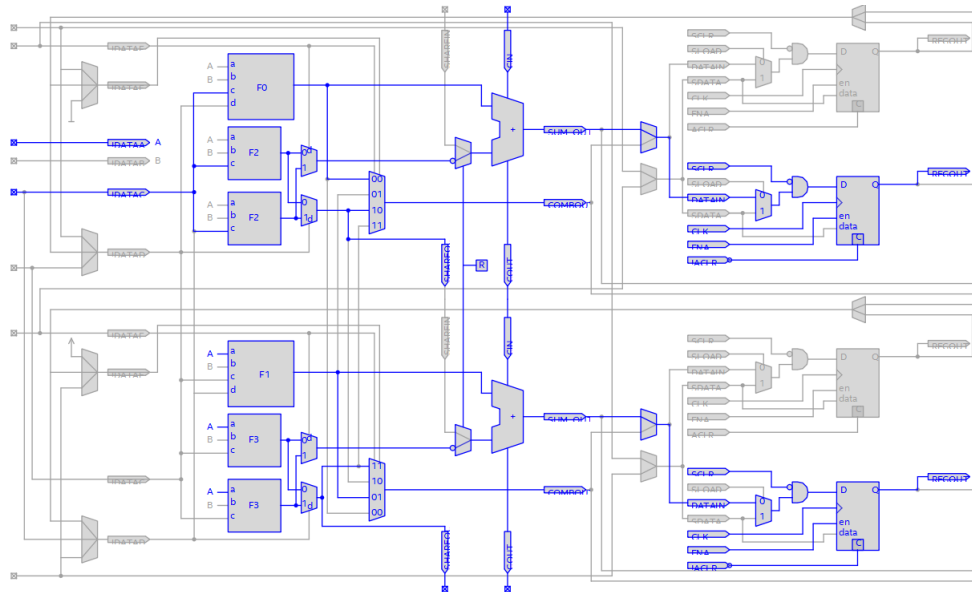
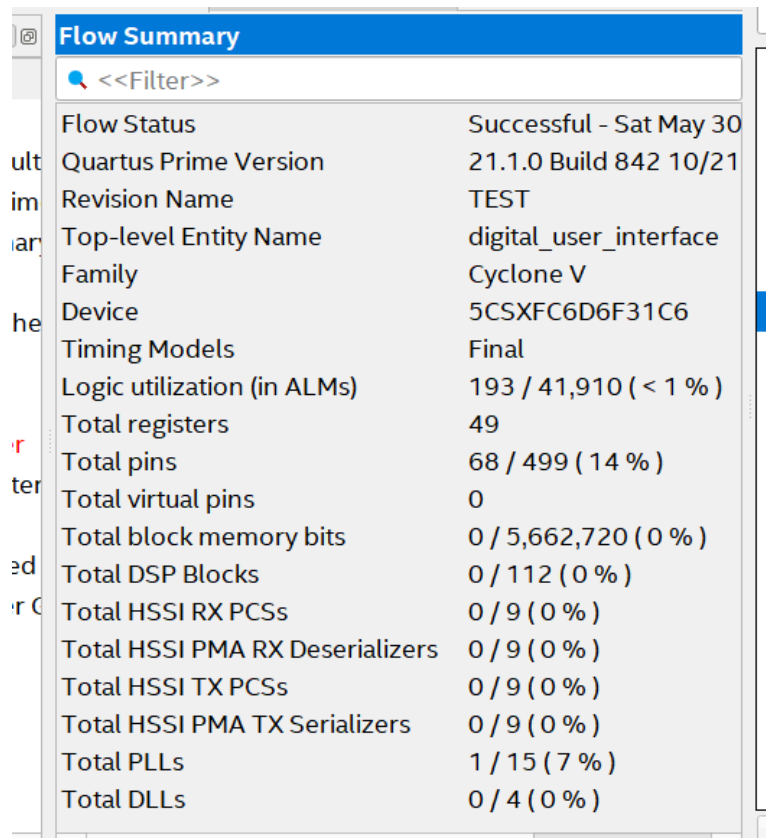


Figure 17: PWM subpart – RTL Viewer: the 16-bit timer feeding the comparison and mode-dependent output logic that produces PWMOut.

5. Full Digital System

Finally the complete digital system (top entity digital_user_interface, revision TEST) was compiled, combining the combinational ALU, the synchronous PWM unit, the I/O wrapper and the PLL. The following reports characterize the whole system.

5.1 Area (logic usage)



Flow Summary	
<<Filter>>	
Flow Status	Successful - Sat May 30
Quartus Prime Version	21.1.0 Build 842 10/21
Revision Name	TEST
Top-level Entity Name	digital_user_interface
Family	Cyclone V
Device	5CSXFC6D6F31C6
Timing Models	Final
Logic utilization (in ALMs)	193 / 41,910 (< 1 %)
Total registers	49
Total pins	68 / 499 (14 %)
Total virtual pins	0
Total block memory bits	0 / 5,662,720 (0 %)
Total DSP Blocks	0 / 112 (0 %)
Total HSSI RX PCSs	0 / 9 (0 %)
Total HSSI PMA RX Deserializers	0 / 9 (0 %)
Total HSSI TX PCSs	0 / 9 (0 %)
Total HSSI PMA TX Serializers	0 / 9 (0 %)
Total PLLs	1 / 15 (7 %)
Total DLLs	0 / 4 (0 %)

Figure 18: Full system – Flow Summary (area).

The complete system uses 193 of 41,910 ALMs (< 1%), 49 registers, and 68 of 499 pins (14%). It uses one of the 15 PLLs (7%) — this is the PLL that divides the 50 MHz board clock down to the working clock. No block memory or DSP blocks are used.

Resource	Usage
Logic utilization (ALMs)	193 / 41,910 (< 1%)
Total registers	49
Total pins	68 / 499 (14%)
Block memory bits	0
DSP blocks	0
PLLs	1 / 15 (7%)

5.2 Maximum frequency (Fmax)

	Fmax	Restricted Fmax	Clock Name	Note
1	214.32 MHz	214.32 MHz	U_PLL alt...ER divclk	

Figure 19: Full system – maximum frequency on the PLL output clock (divclk).

The system maximum frequency on the PLL output clock (U_PLL|altpll_component|...|divclk) is 214.32 MHz. Since the PLL supplies the working clock well below this value, the system has comfortable timing margin (large positive slack).

5.3 Critical (longest) path

Path #1: Setup slack is 495.334		
Path Summary	Statistics	Data Path
Property	Value	
1 From Node	digital_system:U_DIGITAL_SY...Timer:TIMER_INST count_q[1]	
2 To Node	digital_system:U_DIGITAL_S...it_circ:PWM_INST pwm_out_w	
3 Launch Clock	U_PLL altpll_component au...PLL_OUTPUT_COUNTER divclk	
4 Latch Clock	U_PLL altpll_component au...PLL_OUTPUT_COUNTER divclk	
5 Data Arrival Time	7.894	
6 Data Required Time	503.228	
7 Slack	495.334	

Figure 20: Full system – critical (longest) path summary, setup slack 495.334 ns.

The longest path of the full system starts at the internal timer register digital_system|U_DIGITAL_SY...Timer:TIMER_INST|count_q[1] and ends at the PWM output signal digital_system|U_DIGITAL_S...it_circ:PWM_INST|pwm_out_w, both launched and latched on the PLL divided clock. The data arrival time is 7.894 ns against a required time of 503.228 ns, giving a very large setup slack of 495.334 ns (the requirement is relaxed because the working clock is slow).

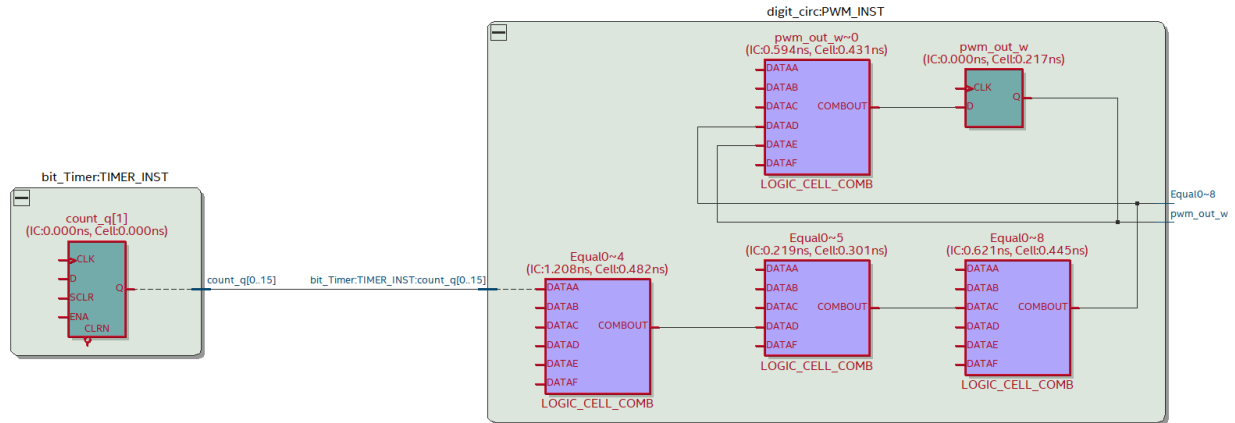


Figure 21: Full system – Technology-Map view of the longest path: timer register → comparison logic cells (Equal0~4/5/8) → PWM output register `pwm_out_w`.

Why this is the frequency-limiting operation

The frequency-limiting operation is located in the synchronous PWM subpart. According to the Timing Analyzer, the critical path starts from the internal timer register `TIMER_INST|count_q[1]` and ends at the PWM output register/signal `PWM_INST|pwm_out_w`. This path includes the propagation of the timer value, the comparison against the X and Y threshold values, and the mode-dependent PWM output control logic. Therefore, the PWM comparison and control logic is the operation that limits the maximum operating frequency of the full digital system: among all the register-to-register paths, the chain “timer → compare-against-X/Y → mode logic → output” has the most combinational stages.

5.4 Shortest path

Path #1: Hold slack is 0.367		
Path Summary Statistics Data Path Waveform Extra Fitter Information		
	Property	Value
1	From Node	digital_system:U_DIGITAL_SY...Timer:TIMER_INST count_q[0]
2	To Node	digital_system:U_DIGITAL_SY...Timer:TIMER_INST count_q[0]
3	Launch Clock	U_PLL altpll_component au...PLL_OUTPUT_COUNTER divclk
4	Latch Clock	U_PLL altpll_component au...PLL_OUTPUT_COUNTER divclk
5	Data Arrival Time	1.845
6	Data Required Time	1.478
7	Slack	0.367

Figure 22: Full system – shortest path (hold check), hold slack 0.367 ns.

The shortest path is the one most critical for the hold check. It is found inside the timer (bit_Timer / TIMER_INST): both its From-Node and To-Node are TIMER_INST|count_q[0], launched and latched by the PLL divided clock. The data arrival time is 1.845 ns and the data-required time is 1.478 ns, giving a small positive hold slack of 0.367 ns.

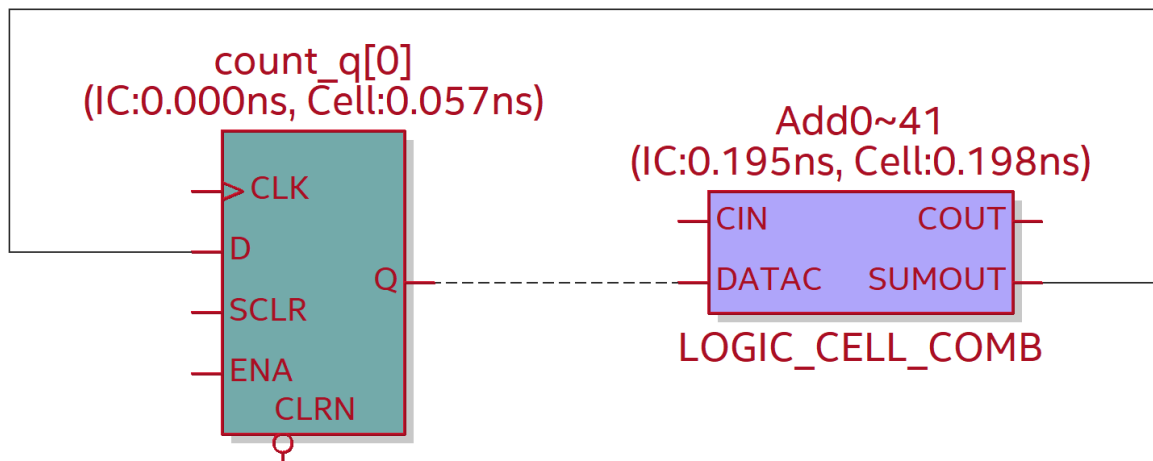


Figure 23: Full system – Technology-Map view of the shortest path: count_q[0] register feeding the Add0~41 logic cell (the timer increment).

Why this is the shortest path

This path is the shortest because count_q[0] is the least-significant bit of the counter. To compute its next value the design only has to feed the increment adder cell (Add0~41) directly — it does not require a long carry chain, unlike the higher counter bits whose next value depends on the carry propagating through all the lower bits. With essentially no interconnect and a single logic cell (mostly a toggle/inversion), the path from the register back to the increment logic is very short, which is exactly what produces the small, hold-critical delay.

5.5 RTL

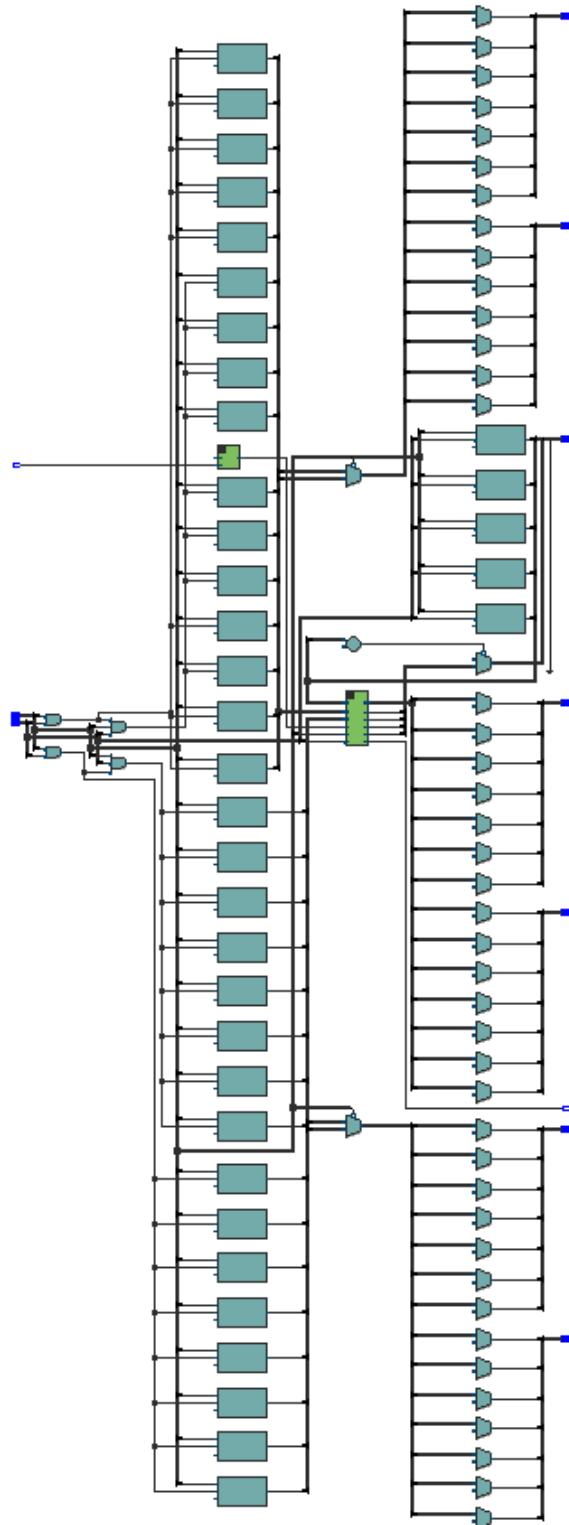


Figure 24: Full digital system – RTL Viewer (timer, registers, comparison and PWM/ALU output logic).

6. PPA Characterization Summary

Area						
	Logic elements (ALMs)	Registers	I/O pins	Embedded memory bits	Embedded 9-bit Multipliers	PLLs
Combinational subpart (ALU)	82	33	35	0	0	0
Synchronous subpart (PWM)	54	49	41	0	0	0

* The ALU register count comes from the timing wrapper (input/output registers); the ALU core itself is purely combinational.

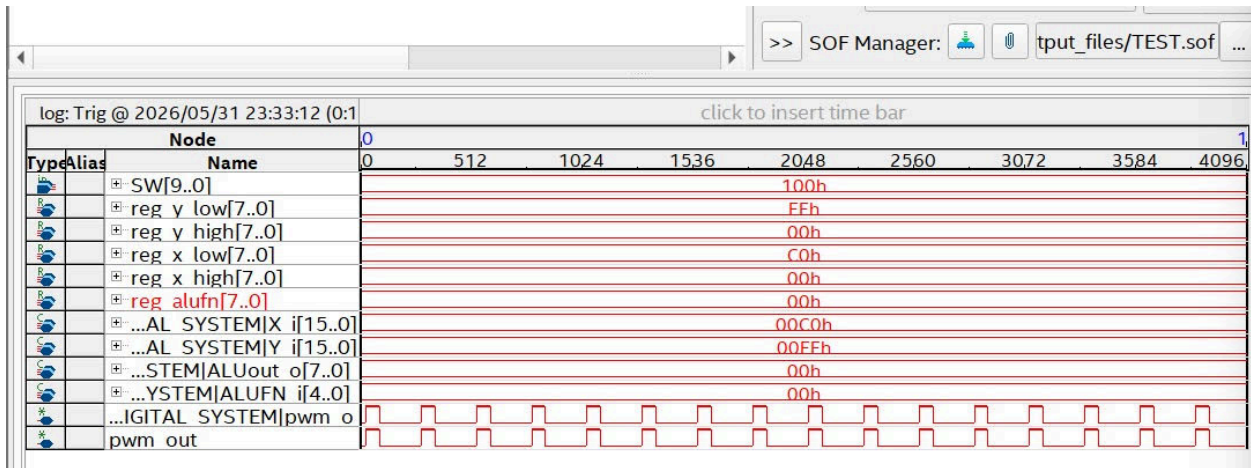
Performance			
	$f_{\max}$	Critical path (what is the slowest submodule and why does it cause the critical path)	$f_{\text{MCLK}} (= f_{\text{sysclk}})$
Combinational subpart (ALU)	155.59 MHz	ALUFN_reg[4] → ALUout_o[7]~reg0. The output-selection multiplexer chain after the arithmetic/shift datapath is the longest path.	None
Synchronous subpart (PWM)	223.21 MHz	count_q[0] → count_q[2] (timer increment / carry). In the full system the limiting path is timer → X/Y compare → PWM mode logic → pwm_out.	214.32 MHz (PLL divclk, full system)

Power				
	Total power consumption	Static power consumption	Dynamic power consumption	I/O power consumption
Combinational subpart (ALU)	424.25 mW	411.25 mW	0.74 mW	12.27 mW
Synchronous subpart (PWM)	422.12 mW	411.24 mW	0.00 mW	10.88 mW

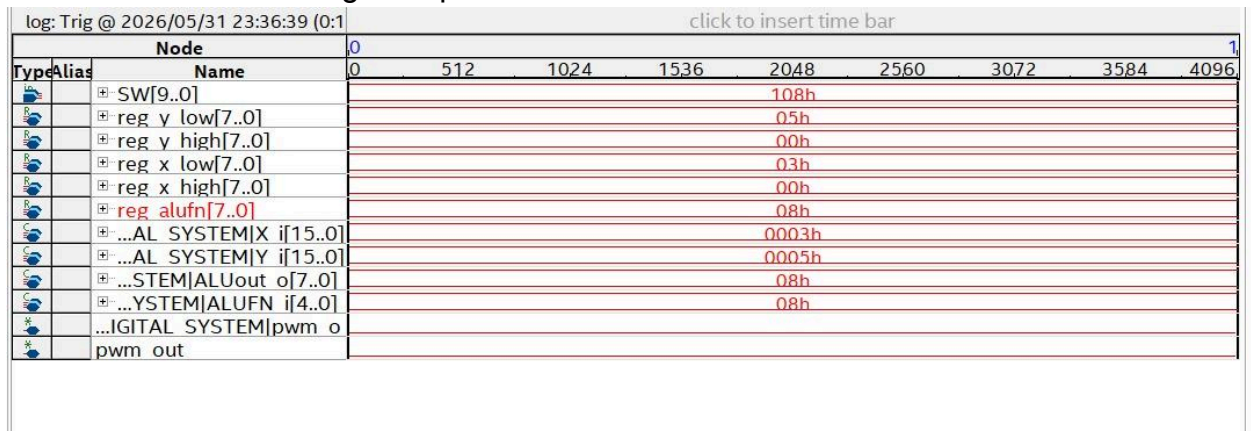
Power Estimation Confidence: Low — insufficient toggle-rate (switching activity) data was provided. The 0.00 mW core dynamic figure for the synchronous subpart is an artifact of this; supplying a simulation .vcd would yield a realistic value. Static power dominates because a small design sits on a large Cyclone V device.

7. signal-tap - verification :

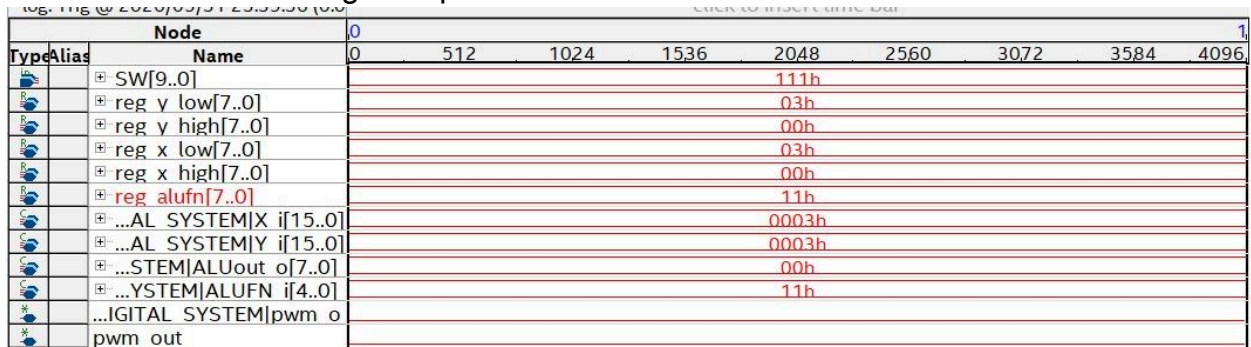
we will demonstrate a signal-tap for an pwm instruction:



we will demonstrate a signal-tap for an alu-add instruction:



we will demonstrate a signal-tap for an alu-shr instruction:



we will demonstrate a signal-tap for an alu-and instruction:

Node		0	1								
Type	Alias	Name	0	512	1024	1536	2048	2560	3072	3584	4096
REG		⊕ SW[9..0]					11Ah				
REG		⊕ reg y low[7..0]					03h				
REG		⊕ reg y high[7..0]					00h				
REG		⊕ reg x low[7..0]					03h				
REG		⊕ reg x high[7..0]					00h				
REG		⊕ reg alufn[7..0]					1Ah				
CPU		⊕ ...AL SYSTEM X i[15..0]					0003h				
CPU		⊕ ...AL SYSTEM Y i[15..0]					0003h				
CPU		⊕ ...STEM ALUout o[7..0]					03h				
CPU		⊕ ...YSTEM ALUFN i[4..0]					1Ah				
MCU		...IGITAL SYSTEM pwm o									
MCU		pwm out									